Polynomials

What is a polynomial?

A **polynomial** in x is an expression like $3x^2 - 5x + 2$, where the powers of x are whole numbers (0, 1, 2, …). The highest power is its **degree**.

Degree	Name	General form	Example
1	Linear	$ax + b$	$2x - 3$
2	Quadratic	$ax^2 + bx + c$	$x^2 - 5x + 6$
3	Cubic	$ax^3 + bx^2 + cx + d$	$x^3 - 1$

ZERO OF A POLYNOMIAL

A number k is a **zero** of p(x) if $p(k) = 0$. For $x^2 - 5x + 6$, $p(2) = 4 - 10 + 6 = 0$, so 2 is a zero (and so is 3).

Seeing zeroes on a graph

The zeroes of p(x) are the **x-coordinates of the points where the graph of y = p(x) meets the x-axis**.

- A linear polynomial (straight line) cuts the x-axis exactly **once**: one zero.
- A quadratic polynomial (a U-shaped **parabola**) can meet the x-axis at **2, 1 or 0** points.
- A cubic polynomial has **at most 3** zeroes.

READING A GRAPH QUESTION

Count where the curve touches or crosses the x-axis. A curve that never meets the x-axis has no real zeroes. A polynomial of degree n has at most n zeroes.

Zeroes and coefficients of a quadratic

If α and β are the zeroes of $ax^2 + bx + c$, then:

RELATIONSHIP BETWEEN ZEROES AND COEFFICIENTS

$\alpha + \beta = -b/a$ and $\alpha\beta = c/a$

FIND THE ZEROES OF $X^2 + 7X + 10$ AND VERIFY THE RELATIONSHIP

Split the middle term: $x^2 + 5x + 2x + 10 = x(x + 5) + 2(x + 5) = (x + 2)(x + 5)$.

Zeroes: **−2 and −5**.

Sum = −7 and $-b/a = -7/1$ ✓. Product = 10 and $c/a = 10/1$ ✓.

Building a quadratic from its zeroes

If you know the sum S and product P of the zeroes, a quadratic polynomial is:

QUADRATIC WITH ZERO-SUM S AND ZERO-PRODUCT P

$k(x^2 - Sx + P)$, for any non-zero constant k

SUM OF ZEROES = −3, PRODUCT = 2

$x^2 - (-3)x + 2 =$ **$x^2 + 3x + 2$**.

WATCH THE SIGN OF B

The sum of zeroes is **$-b/a$**, not b/a. For $2x^2 - 8x + 6$, the sum is $-(-8)/2 = 4$.

Cubic polynomials (for reading)

For $ax^3 + bx^2 + cx + d$ with zeroes α, β, γ: $\alpha + \beta + \gamma = -b/a$, $\alpha\beta + \beta\gamma + \gamma\alpha = c/a$ and $\alpha\beta\gamma = -d/a$.

Quick recap

- Degree = highest power. Zeroes = where the graph meets the x-axis.
- A degree-n polynomial has at most n zeroes.
- Quadratic: $\alpha + \beta = -b/a$, $\alpha\beta = c/a$.
- Quadratic from zeroes: $x^2 - (\text{sum})x + (\text{product})$.

Practice questions

1. The zeroes of $x^2 - 2x - 8$ are
   (a) 2 and –4 (b) 4 and –2 (c) –4 and –2 (d) 4 and 2
2. The graph of y = p(x) meets the x-axis at 3 points. The number of zeroes of p(x) is
   (a) 1 (b) 2 (c) 3 (d) cannot be found
3. A quadratic polynomial whose sum of zeroes is 4 and product is 1 is
   (a) $x^2 + 4x + 1$ (b) $x^2 - 4x + 1$ (c) $x^2 - x + 4$ (d) $4x^2 + x - 1$
4. For $3x^2 - 6x + 9$, the sum of the zeroes is
   (a) –2 (b) 2 (c) 3 (d) –3
5. The maximum number of zeroes a cubic polynomial can have is
   (a) 1 (b) 2 (c) 3 (d) 4
6. If one zero of $x^2 - 5x + k$ is 2, then k equals
   (a) 4 (b) 6 (c) –6 (d) 10

Answers: 1. b 2. c 3. b 4. b 5. c 6. b